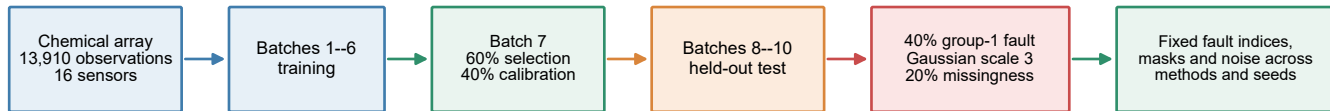
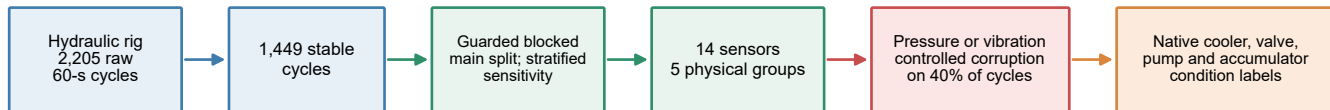


Experimental protocol, fixed interventions and evidence boundary



Temporal acquisition order over 36 months; no test batch enters training, checkpoint selection or calibration



Physical component labels are native to the rig; sensor faults in both datasets remain controlled interventions